opposite reason— because it has no effect on bodies …

I see the distinction that Gould is making, but I cannot see it as a fundamental one. On the contrary, I still see selfish DNA as a special case of the whole theory of the selfish gene, which is precisely how the idea of selfish DNA originally arose. (This point, that selfish DNA is a special case, is perhaps even clearer on page 237 of this book than in the passage from page 57 cited by Doolittle and Sapienza, and Orgel and Crick. Doolittle and Sapienza, by the way, use the phrase 'selfish genes', rather than 'selfish DNA', in their title.) Let me reply to Gould with the following analogy. Genes that give wasps their yellow and black stripes increase in frequency because this ('warning') colour pattern powerfully stimulates the brains of other animals. Genes that give tigers their yellow and black stripes increase in frequency 'for precisely the opposite reason'— because ideally this (cryptic) colour pattern does not stimulate the brains of other animals at all. There is indeed a distinction here, closely analogous (at a different hierarchical level!) to Gould's distinction, but it is a subtle distinction of detail. We should hardly wish to claim that the two cases 'could not be more different in the structures of explanation that nurture them'. Orgel and Crick hit the nail on the head when they make the analogy between selfish DNA and cuckoo eggs: cuckoo eggs, after all, escape detection by looking exactly like host eggs.

Incidentally, the latest edition of the *Oxford English Dictionary* lists a new meaning of 'selfish' as 'Of a gene or genetic material: tending to be perpetuated or to spread although of no effect on the phenotype.' This is an admirably concise definition of 'selfish DNA', and the second supporting quotation actually concerns selfish DNA. In my opinion, however, the final phrase, 'although of no effect on the phenotype', is unfortunate. Selfish genes *may* not have effects on the phenotype, but many of them do. It would be open to the lexicographers to claim that they intended to confine the meaning to 'selfish DNA', which really does have no phenotypic effects. But their first supporting quotation, which is from *The Selfish Gene*, includes selfish genes that do have phenotypic effects. Far it be from me, however, to cavil at the honour of being quoted in the *Oxford English Dictionary*!

I have discussed selfish DNA further in *The Extended Phenotype* (pp. 156–64).